Assignment for Week 2

Author: Grace Zeng

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install.packages("haven") library(dplyr, warn.conflicts = FALSE) library(ggplot2) library(nycflights13) library(MASS) knitr::opts_chunk$set(echo = TRUE) set.seed(100)

"'

Package Loading

```
library(dplyr, warn.conflicts = FALSE)
library(ggplot2)
library(haven)
```

Assignment

1) 5.2.1 (p. 87 **Pemberton and Rau**)
   Answer:
   $\sum_{n=1}^{100} n = (1 + 100) \times 100 \div 2 = 5050$
2) 5.4.2 (p. 98 **Pemberton and Rau**)
   Answer:

(a) Yes, $f(x)$ is continuous at $x = 0$, because $\lim_{x\to 0} f(x) = f(0) = 0$.
(b) Yes, $f(x)$ is continuous at $x = 1$, because $\lim_{x\to 1} f(x) = f(1) = 1$.
However, $f(x)$ is not a continuous function, because $\lim_{x\to 2} f(x)$ does not exist.

3) 6.3.2 (p. 114 **Pemberton and Rau**)
   Answer:

(a) $(4x - 2)x^3 = 4x^4 - 2x^3$, and so $((4x - 2)x^3)' = 16x^3 - 6x^2$.
(b) $x(x + 1)(X + 2) = x^3 + 3x^2 + 2x$, and so $(x(x + 1)(X + 2))' = 3x^2 + 6x + 2$.
(c) $\frac{3x^2+1}{2x} = \frac{3x}{2} + \frac{1}{2x}$, and so $(\frac{3x^2+1}{2x})' = \frac{3}{2} - \frac{1}{2x^2}$.
(d) $\frac{(2x-1)(x+1)}{x} = 2x + 1 - \frac{1}{x}$, and so $(\frac{(2x-1)(x+1)}{x})' = 2 + \frac{1}{x^2}$.
(e) $(x + \frac{a}{x})(x - \frac{a}{x}) = x^2 - \frac{a^2}{x^2}$, and so $((x + \frac{a}{x})(x - \frac{a}{x}))' = 2x + \frac{2a^2}{x^3}$.
(f) $\frac{x-a}{bx^3} = \frac{1}{bx^2} - \frac{a}{bx^3}$, and so $(\frac{x-a}{bx^3})' = -\frac{2}{bx^3} + \frac{3a}{bx^4}$.

4) 6.3.3 (p. 114 **Pemberton and Rau**)
   Answer:
   $f'(x) = 4x^2 + 3 \geq 3 > 0$ for all x. Therefore, $f(x)$ is a monotonically increasing function on $\mathbb{R}$, and so the equation $x^5 + 3x = 12$ has one and only one solution on $\mathbb{R}$, which is between 1 and 2.

5) Using the world religion dataset, what country experience the largest percentage increase in Islam over the dataset? Answer:

```
religionchange <- religions
                  %>% group_by(country)
                  %>% summarise(maxinc = max(c(diff(isgenpct), 0), na.rm = T),
                                maxdec = min(c(diff(isgenpct), 0), na.rm = T))
                  %>% arrange(maxdec)

religionchange[1,1]
# A tibble: 1 × 1
  country
    <chr>
1   Qatar
```

Therefore, the country that experiences the largest percentage increase in Islam over the dataset is Qatar.

6) Without using R, how many rows and columns does df1, df2, and df3 have?

```
dfA <- data.frame(x = 0:10, yA = rnorm(11,-1))
dfB <- data.frame(x = 5:15, yB = rnorm(11, 1))

df1 <- left_join(dfA, dfB)
df2 <- na.omit(df1)
df3 <- anti_join(dfA, dfB)
```

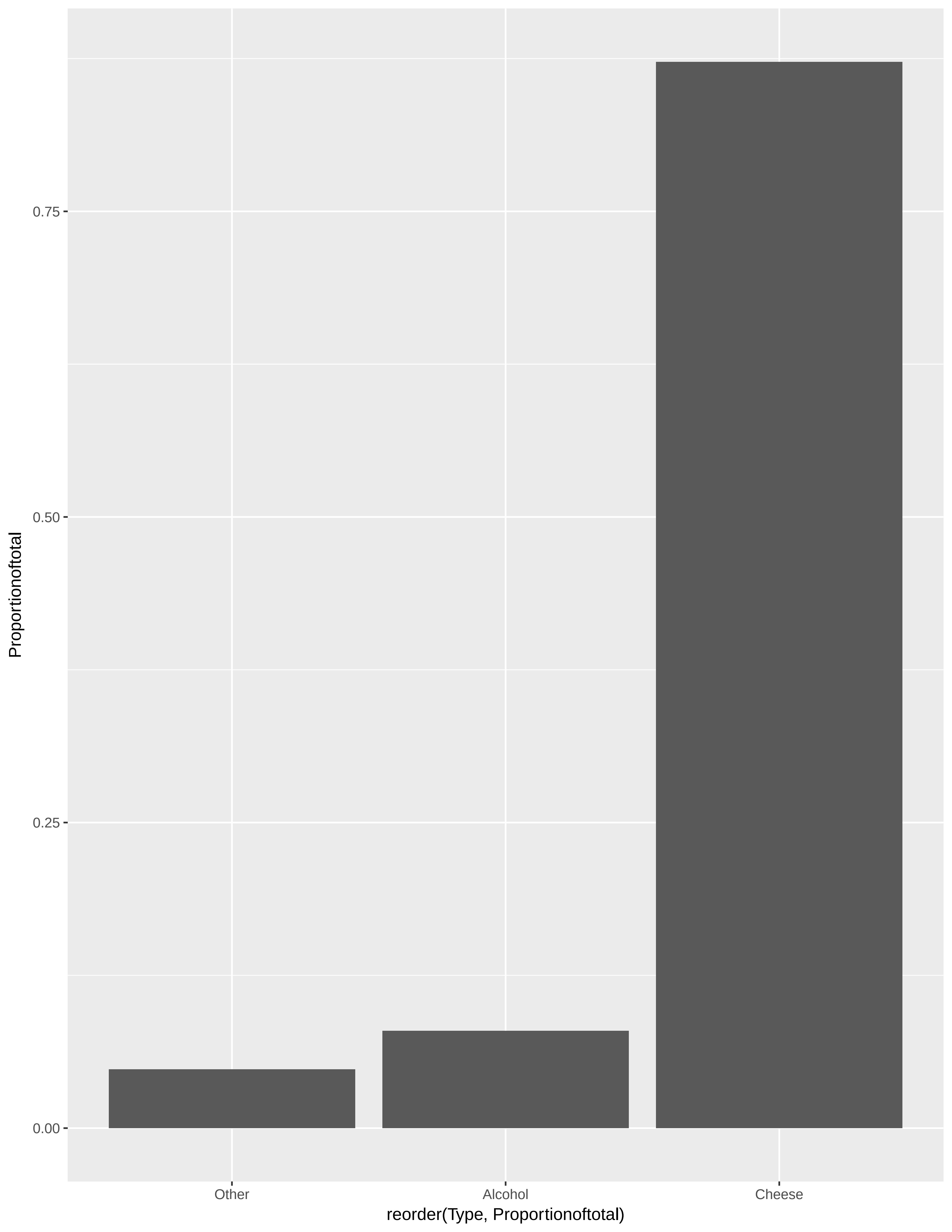
Answer:
df1 has 11 rows and 3 columns;
df2 has 6 rows and 3 columns;
df3 has 5 rows and 2 columns.

7) Turn the following code into a beautify dplyr piped code. Answer:

```
dfFondue <- data.frame(Ingredient=c("GruyÃ¨re", "vacherin", "Kirsch","Garlic",
            "White Wine", "cornstarch"))
dfFondueCatalogue <- dfFondue
                    %>% left_join(mutate(dfFondue, Origin = c("GruyÃ¨res", "Fribourg",
                                    "ZÃ¼rich", "Lucern", "Jura", "Iowa")))
                    %>% left_join(mutate(dfFondue, Price = c(12.99, 14.99, 15.00, 1.50,
                                    12.00, 0.90)))
                    %>% left_join(mutate(dfFondue, Amount = c(1, 1, .01, 1, .2, .05)))

dfFondueCatalogue$Proportionoftotal <- dfFondueCatalogue$Price*dfFondueCatalogue$Amount/
                                    sum(dfFondueCatalogue$Price*dfFondueCatalogue$Amount)

ggplot(dfFondueCatalogue, aes(reorder(Ingredient,Proportionoftotal),
                                    Proportionoftotal))+geom_col()
```

8) Create a stacked barplot of the Fondue data with the incredients sorted into alcohol (Wine and Kirsch), Cheese (Gruyere and Vacherin) and Other. Make sure the labels are correct.

```
dfFondueCatalogue <- dfFondue
  %>% left_join(mutate(dfFondue, Origin = c("GruyÃ¨res", "Fribourg",
                      "ZÃ¼rich", "Lucern", "Jura", "Iowa")))
  %>% left_join(mutate(dfFondue, Price = c(12.99, 14.99, 15.00,
                      1.50, 12.00, 0.90)))
  %>% left_join(mutate(dfFondue, Amount = c(1, 1, .01, 1, .2, .05)))
  %>% left_join(mutate(dfFondue, Type = c("Cheese", "Cheese",
                      "Alcohol", "Other", "Alcohol", "Other")))

dfFondueCatalogue$Proportionoftotal <-
        dfFondueCatalogue$Price*dfFondueCatalogue$Amount/
        sum(dfFondueCatalogue$Price*dfFondueCatalogue$Amount)

ggplot(dfFondueCatalogue, aes(reorder(Type,Proportionoftotal),
        Proportionoftotal))+geom_col()
```



0.75
0.50
0.25
0.00
Proportionoftotal
Other
Alcohol
Cheese
reorder(Type, Proportionoftotal)